

R for marine ecologists: wrangling Earth System Model outputs

Jessica Bolin

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Preface

Pilot workshop date: 06/03/2025

Location: Lecture Hall, Bodega Marine Laboratory

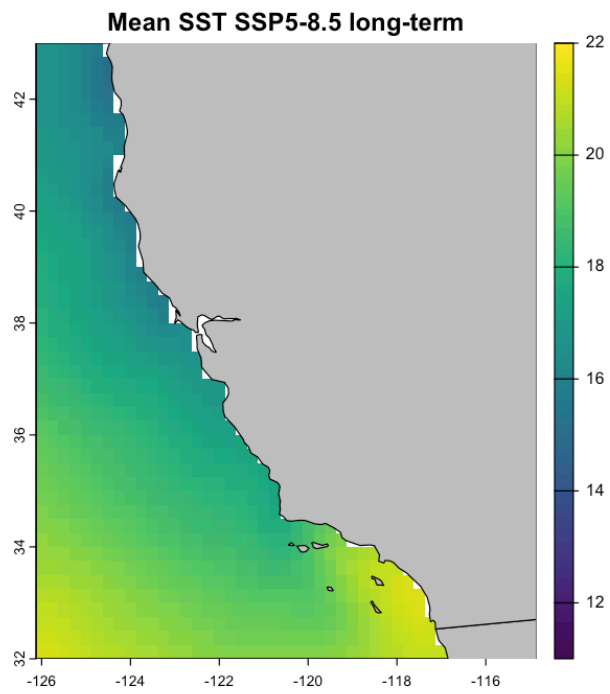
Instructors: Jessica Bolin, Mikaela Provost, Nerea Lezama-Ochoa and Mer Pozo Buil

R is undoubtedly the programming language of choice for many marine ecologists. However, navigating the complexities of downloading, processing and wrangling gridded oceanographic data within R can represent a steep learning curve. As climate-related projects continue to receive funding and the impacts of climate variability and change on our oceans become increasingly evident, **the ability to efficiently interact with, and use, Earth System Model (ESM) outputs for our research is more critical than ever.**

The aim of our one-day R course is to train students, researchers and faculty in skills needed to be proficient in creating projections of physical variables from ESM outputs, with the goal of using them for ecological applications. **Specifically, we'll be creating projections of sea-surface temperature across two time periods out to 2100, across two climate scenarios, for the California Current region.**

Our workshop is designed around Step 8 (pre-process the data) in the workflow of best practices for using ESMs for marine ecologists, as defined in Figure 1 within (Schoeman et al. 2023).

Our workshop is supported by CMSI, and is hosted by Bodega Marine Lab. We thank both organisations for their funding and support.



Acronyms

We will be using **a lot** of acronyms during today's workshop..!

- ACCESS-CM2 = Australian Community Climate and Earth System Simulator Coupled Model Version 2
- CD0 = Climate Data Operators
- CMIP6 = Coupled Model Intercomparison Project Version 6
- ESGF = Earth System Grid Federation
- ESM = Earth System Model
- GCM = Global Coupled Model
- IPCC = Intergovernmental Panel on Climate Change
- IPSL-CM6A-LR = Institut Pierre-Simon Laplace Climate Model for Phase 6 of CMIP - Low Resolution
- netCDF = Network Common Data Form (file format = `.nc`)
- OISST = Optimum Interpolation Sea Surface Temperature
- omon = monthly time step
- wget = World Wide Web Get
- SSPs = Shared socio-economic pathways
- SST = Sea-surface temperature
- tos = sea-surface temperature (temperature ocean surface)

About

Welcome to the workshop! This workshop is designed to assist you in getting started with using Earth System Model (ESM) outputs using R.

The workshop will run from 10 am to 5 pm (PST) on Tuesday 3rd June, 2025.

Instructors

Today's instructors are **Jessica Bolin**, **Mikaela Provost**, **Mer Pozo Buil** and **Nerea Lezama-Ochoa**.

Jessie is a Postdoc in Provost Lab, working on modeling and mapping climate refugia for red abalone across California. Jessie has a background in quantitative marine science and obtained her PhD from the University of the Sunshine Coast in 2024, and has broad research interests spanning climate adaptation, fisheries forecasting, marine disease ecology and sustainable seafood. Jessie has been using R since 2016, and has worked with ESMs since 2021. jabbolin@ucdavis.edu

Mikaela is an Assistant Professor at UC Davis, and the leader of Provost Lab. Mikaela's research uses theoretical and empirical approaches to address applied problems in fisheries management and marine conservation. Her projects span multiple species and places, and the methods I use can be adapted for a wide variety of taxa. mmprovost@ucdavis.edu

Mer is an Associate Project Scientist at the University of California Santa Cruz. Mer's research interests are in ocean modeling and large-scale circulation, specifically about subsurface ocean variability, large-scale changes in water masses and circulation, and how they affect upwelling systems.

Nerea is an Associate Project Scientist at the University of California Santa Cruz. Nerea is currently working on the conservation and management of species taken incidentally in tropical tuna purse seine fisheries (bycatch). Nerea's main research interests are ecological modeling, spatial ecology, oceanography and ecosystem-based management approaches for the conservation of marine top predators.

Chapter index

1. What is an ESM?

1. Experiments
2. Climate scenarios (SSPs)

2. Download ESM output from ESGF

1. The ESGF database
2. Download `wget` shell scripts
3. Download `.nc` files with `wget` scripts

3. Remapping

1. Inspect an ESM
2. Remapping with CDO and R

4. OISST / observed data

1. Download OISST
2. Preprocess OISST
3. Visualize OISST

5. Bias correction & downscaling

1. OISST and ESM climatologies
2. ESM anomalies
3. Interpolate ESM anomalies to OISST grid
4. Bias correction

6. Evaluate accuracy of historical ESM projections

1. Brief explainer
2. Create baseline climatologies
3. Taylor Diagrams

7. Making projections

1. Time series
2. Spatial projections
3. Uncertainty

Workshop materials

The workshop data are ~2.5 GB, so ensure you have enough storage space on your machine before downloading, or bring an external SSD/Hard-drive.

If you're a Github user, you can fork and clone the repo [here](#). Then, you will then need to download the workshop `__data` folder [here](#), which needs to be copied into the root directory of `esmRworkshop_notes`. We've added the `/__data` folder to the `.gitignore` file within the root directory, since we'll be working with large files (and Github breaks when pushing/pulling large files).

If you don't use Github, you can instead download the complete workshop repository [here](#).

Alternatively, you can obtain the materials on the day from us via USB.

On the day, we'll be working with code contained in the `__scripts` directory.

We have inserted the expected time duration for running processor-intensive chunks of code in **blockquotes** throughout the eBook. This is based on Jessie's machine (MacBook Pro 2023 M3 Max 64 GB Memory). Your machine may be faster or slower, depending on its specifications.

What to bring

- Laptop (highly, highly, **highly** recommend macOS) with internet connection. **Please ensure you have administrator permissions** - consult with your IT department on how to do this.
- Install the necessary software, listed on the **Prerequisites** page.

Expected outcomes

By the end of the course, participants should be able to:

- Download, inspect, and wrangle netCDF file formats within R
- Use `CD0` to speed up netCDF file manipulation
- Download ESM output from ESGF and understand terminology re. variables/naming conventions
- Remap and bias-correct ESM outputs from the CMIP6 suite across multiple climate scenarios and time periods
- Make publication-quality figures

Transportation to/from UC Davis

We are arranging for a UCD minivan to transport participants to/from UC Davis main campus. The van will be driven by Mary Fisher and has space for up to 12 participants - first-in, first-served. Sign up sheet TBA.

Food

Lunch (sandwiches) will be provided, with vegan and vegetarian options available.

Acknowledgements

We acknowledge funding from the UC Davis Coastal and Marine Science Institute's internal grant scheme (2024-2025), and thank Bodega Marine Laboratory for providing space and equipment to run the workshop.

We are very thankful for Barb Muhling (UCSC) and Alice Pidd (UniSC) who reviewed workshop materials. We also thank Isaac Brito-Morales for developing the following R workshop materials [here](#) and [here](#) on analysing netCDFs and climate data, which our workshop took inspiration from. We also thank Dave Schoeman who developed and shared the first iteration of some code in this workshop, and who introduced Jessie to the world of climate models in the first place!

About this book

This is a Quarto book. To learn more, click [here](#).

Prerequisites

Prerequisites

! Workshop materials

The workshop data are ~2.5 GB, so ensure you have enough storage space on your machine before downloading, or bring an external SSD/Hard-drive.

If you're a Github user, you can fork and clone the repo [here](#). Then, you will then need to download the workshop `__data` folder [here](#), which needs to be copied into the root directory of `esmRworkshop_notes`. We've added the `/__data` folder to the `.gitignore` file within the root directory, since we'll be working with large files (and Github breaks when pushing/pulling large files).

If you don't use Github, you can instead download the complete workshop repository [here](#).

On the day, we'll be working with code contained in the `__scripts` directory.

Our course is not intended for absolute beginners in R. While participants don't need to be highly proficient, a basic understanding of R and programming syntax is required. For example, participants should know how to create an R project and open a script, read in data using commands, make basic plots using `base` R, and write for loops. A basic understanding of the shell/command line is advantageous, but not required. Further, previous experience using the `terra` package and working with `netCDF` files is a plus, but again, not required.

In addition to having R and RStudio already installed, you will need to install (i) `CD0`, (ii) `NCO`, (iii) `wget`, (iv) `Panoply` and (v) a suite of R packages.

! Important

We **strongly** recommend bringing a macOS laptop to the workshop. We've found that Windows users often run into issues with installing and using `CD0`, which is required for the workshop (see below). If you have a Windows laptop, you're still welcome to attend, but the instructors will be using macOS during the workshop. As such, we're unlikely to be able to fix any Windows-related errors or software installation problems (and we unfortunately can't spend time troubleshooting any Windows issues).

To see if you may have software issues during our workshop, install the prerequisites

below, download this folder [here](#), and run the `runme.R` script. If you don't run into any errors, great! If you do, you will need to do some troubleshooting before the workshop.

CDO

You will need **CDO** (Climate Data Operators) installed prior to the workshop. CDO comprises a suite of over 600 operators for standard (and speedy) processing of climate and forecast model outputs. More information on CDO found [here](#).

MacOS

The simplest way to install CDO is via Homebrew. If you haven't already installed Homebrew, do so [here](#). Then, open terminal and run the following code:

```
#### In Terminal  
  
brew install cdo
```

More information available on the [CDO MacOS website](#).

Windows

CDO is meant for use on POSIX-compatible operating systems (e.g., like Linux and MacOS), so downloading on Windows requires some extra steps.

Recent versions of Windows (≥ 10) includes an Ubuntu embedded Linux, offering the opportunity to install CDO via Ubuntu's native package manager. First, install the Ubuntu app from the Microsoft Store application. Then open the Ubuntu terminal and type:

```
#### In Terminal  
  
sudo apt-get upgrade  
sudo apt-get install cdo #write your password, if prompted
```

More information available on the [CDO Windows website](#).

NCO

Similar to CDO, we will install [NCO \(netCDF operators\)](#), which are a suite of operators that take netCDF file formats and facilitate file manipulation.

Mac

```
#### In Terminal
```

```
brew install nco
```

Windows

Installing NCO on windows can be tricky. [Here](#) is a discussion forum.

wget

wget is a bash command used for downloading files from the internet. We will use this to download our ESMs via **wget** scripts from the ESGF website.

macOS

Run the following in terminal:

```
#### In Terminal
```

```
brew install wget
```

Windows

Check out the link [here](#). You may need to do some extra fiddling - see [here](#) or check out Youtube for further help.

Panoply (optional)

[Panoply](#) is a data viewer for netCDF, HDF and GRIB data arrays, administered by NASA Goddard Space Flight Center. **Panoply requires that your computer has a compatible Java 11 (or later version) JRE or JDK installed.**

Download Panoply [here](#).

! Important

Panoply is **optional**. All coding exercises can be completed without it, but it provides a convenient way to inspect and visualize netCDF files.

R Packages

Ensure you have the following R packages installed, listed below.

```
#### In R

#Install pacman
if (!require("pacman")) install.packages("pacman")

# Install packages, if not already installed
pacman::p_load(tidyverse, # Working with 'tidy' data
               furr, parallelly, purrr, # Run functions in parallel
               tictoc, beeper, # Understand code execution time
               ncdf4, raster, terra, # For working with netCDFs
               tmap, maps, viridis, # Map visualization
               plotrix, # Taylor Diagrams
               RCurl, xml2, rvest, # Downloading files
               zoo)
```

Changing your path

Lastly, once you have downloaded the repository, you will need to open `__scripts/helpers.R`, and change your `pth` object to suit your machine (see Line 7). I've set mine as `"/Users/admin/Documents/GitHub/esmRworkshop_website"`; change yours to reflect where this folder is located on your machine.

! Below are the dependencies and package versions the instructors will use for the workshop.

In R

```
> devtools::session_info()
```

```
Session info
setting  value
version  R version 4.4.1 (2024-06-14)
os       macOS Sonoma 14.5
system   aarch64, darwin20
ui       RStudio
language (EN)
collate  en_US.UTF-8
ctype    en_US.UTF-8
```

Schedule

1 What is an ESM?

i Learning objective:

1. Introduction to ESMs (theory)

- Describe what Earth System Models (ESMs) are, how they work, and their role in marine ecological research.
- Explain the CMIP6 framework, climate scenarios, and considerations for selecting appropriate climate models.

They use mathematical laws and equations

i Model jargon

You may have heard the terms climate model, Earth System model (ESM), general circulation model (GCM) and coupled atmosphere-ocean general circulation models (AOGCMs). These are not all interchangeable! Historically, a ‘climate model’ referred to AOGCMs or GCMs, which do not simulate biogeochemical processes. ESMs, on the other hand, evolved from GCMs to simulate ocean chemistry and marine plankton due to their importance in carbon cycling. As such, ESMs simulate the 3D evolution of the ocean and its biogeochemical processes.

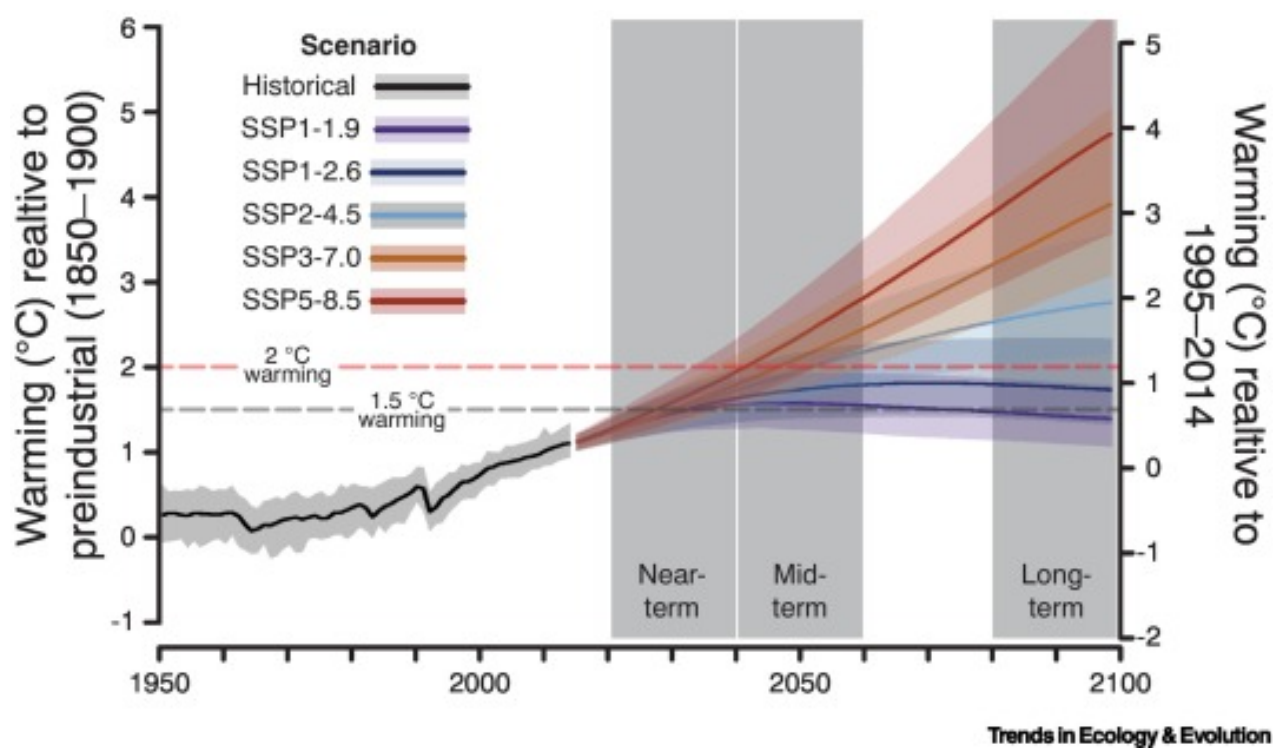
~100 distinct climate models

49 different modeling groups

1.1 Experiments

[here](#)

1.2 Climate scenarios (SSPs)



1.3 ESMs we're using today!

2 Download ESM output from ESGF

i Learning objective:

2. Downloading ESM output (theory and live coding)
 - Access and download CMIP6 model output from the Earth System Grid Federation (ESGF).
 - Use shell scripts and R to efficiently download and manage large climate datasets.

2.1 Downloading CMIP6 Earth System Models

2.2 Download wget shell scripts

2.2.1 Navigate to the Metagrid node

- [here](#)
- [nodes](#)
- [LLNL Metagrid](#)

2.2.2 Filtering ESM components

-
-
-
-

Select a Project

CMIP6

CMIP6 Website

Filter By Transfer Options

☐ Any ☒ Only Globus Transferrable

Collapse All

Filter with Facets

General

Activity ID : CMIP (1) ×

Data Node : Select option(s)

Identifiers

Source ID : ACCESS-CM2 (1) ×

Institution ID : Select option(s)

Source Type : Select option(s)

Experiment ID : historical (1) ×

Sub Experiment ID : Select option(s)

Resolutions

Labels

Variant Label : r1i1p1f1 (1) ×

Grid Label : Select option(s)

Classifications

Table ID : Select option(s)

Frequency : mon (1) ×

Realm : Select option(s)

Variable ID : tos (1) ×

2.2.3 Download wget shell script

Home

1 results found for CMIP6

Query String: latest = true AND (variable_id = tos) AND (frequency = mon) AND (variant_label = r1i1p1f1) AND (source_id = ACCESS-CM2) AND (experiment_id = historical) AND (data_node = aims3.llnl.gov OR esgf-data1.llnl.gov OR esgf-data2.llnl.gov)

tos x mon x r1i1p1f1 x ACCESS-CM2 x historical x aims3.llnl.gov x esgf-data1.llnl.gov x esgf-data2.llnl.gov x Clear All

☐ Cart	Dataset ID	Files	Total Size	Version	Download Options	Globus Ready
☐ + ☐	CMIP6.CMIP.CSIRO-ARCCSS.ACCESS-CM2.historical.r1i1p1f1.Omon.tos.gn	1	466.53 MB	20191108	wget Download	☒

< 1 > 10 / page

2.2.4 Rinse and repeat

2.3 Download .nc files with wget scripts

Warning: storage

In total, we are about to download eight .nc files that comprise 2.65 GB of data. These are just for two models and two scenarios + historical, at a monthly time-step. Storing ESM outputs can take a lot of space - for example, if you're working with daily data across multiple models and climate scenarios, you'll almost certainly need an external hard drive or other storage solution.

2.3.1 Define dependencies

```
source(paste0(getwd(), "/__scripts/helpers.R"))
```

2.3.2 Prepare for parallel processing

```
parallelly::availableCores() # Output says I have 16 cores on my machine  
w <- 14 # Leave two cores for other background processes
```

2.3.3 Function to download files

```
wget_files <- function(script) {  
  setwd(paste0(pth, cmip_pth)) #set directory to where data will be stored  
  system(paste0("bash ", script, " -s")) #run wget on the shell script  
  setwd(pth) #set directory back to home directory  
}
```

```
files <- list.files(paste0(pth, wget_pth), #full path where wget scripts are stored  
  pattern = "wget", #only files with wget in the name  
  full.names = TRUE #list the full path  
)  
files
```


2.3.4 Run function in parallel

```
plan(multisession, workers = w) # Change to multi-threaded processing
tic(); future_walk(files, wget_files); toc() #Run the function in parallel
plan(sequential) # Return to single threaded processing (i.e., sequential/normal)
```

Download speeds

Download speeds depend on your Wifi connection, the capabilities of your local machine, and the performance of the server/node you're attempting to connect to. You can check whether ESGF servers are online [here](#).

```
list.files(paste0(pth, cmip_pth))
```

```
system(paste0("rm ", pth, cmip_pth, "/",  
              "tos_Omon_ACCESS-CM2_ssp585_r1i1p1f1_gn_210101-230012.nc"))  
system(paste0("rm ", pth, cmip_pth, "/",  
              "tos_Omon_IPSL-CM6A-LR_ssp585_r1i1p1f1_gn_210101-230012.nc"))
```

Beware of rm

rm is a handy function for removing files, but there is no way of retrieving them if you've accidentally removed the wrong file. So, use this function wisely and make sure you're 100% confident you're removing the correct file!

3 Remapping

i Learning objective:

3. Regridding and statistically downscaling (live coding)

- Explain why regridding and statistical downscaling are required for ESMs.
- Apply regridding and statistical downscaling methods to ESM outputs in R.

! Important

Note that interpolation, regridding and remapping mean the same thing (for the purposes of our workshop).

3.1 Inspect an ESM

3.1.1 Panoply

Panoply — Sources

Create Plot Combine Plot Open Remove Remove All Hide Info

Datasets Catalogs Bookmarks

Name	Long Name	Type
tos_Omon_ACCESS-CM2_historical...	ACCESS-CM2 output prepared for CMIP6	Local File
i	cell index along first dimension	1D
j	cell index along second dimension	1D
latitude	latitude	Geo2D
longitude	longitude	Geo2D
time	time	1D
time_bnds	time bnds	2D
tos	Sea Surface Temperature	Geo2D
vertices_latitude	vertices latitude	Geo2D
vertices_longitude	vertices longitude	Geo2D

Show: All variables

File
"tos_Omon_ACCESS-CM2_historical_r1i1p1f1_gn_185001-201412.nc"
File type: Hierarchical Data Format, version 5

```
netcdf /Users/admin/Documents/GitHub/BMLworkshop/data/tos_Omon_ACCESS-CM2_
dimensions:
  time = UNLIMITED; // (1980 currently)
  j = 300;
  i = 360;
  bnds = 2;
  vertices = 4;
variables:
  double time(time=1980);
  :bounds = "time_bnds";
  :units = "days since 1850-01-01";
  :calendar = "proleptic_gregorian";
  :axis = "T";
  :long_name = "time";
  :standard_name = "time";
  :ChunkSizes = 10; // uint
  double time_bnds(time=1980, bnds=2);
  :ChunkSizes = 10, 20; // uint
  int j(j=300);
  :units = "1";
  :long_name = "cell index along second dimension";
  int i(i=360);
  :units = "1";
  :long_name = "cell index along first dimension";
  double latitude(j=300, i=360);
  :standard_name = "latitude";
  :long_name = "latitude";
  :units = "degrees_north";
  :missing_value = 1.0E20; // double
```

Panoply — Sources

Create Plot Combine Plot Open Remove Remove All

Datasets Catalogs Bookmarks

Name	Long Name	Type
tos_Omon_ACCESS-CM2_historical...	ACCESS-CM2 output prepared for CMIP6	Local File
i	cell index along first dimension	1D
j	cell index along second dimension	1D
latitude	latitude	Geo2D
longitude	longitude	Geo2D
time	time	1D
time_bnds	time bnds	2D
tos	Sea Surface Temperature	Geo2D
vertices_latitude	vertices latitude	Geo2D
vertices_longitude	vertices longitude	Geo2D

Variable "tos"

In file "tos_Omon_ACCESS-CM2_historic

```
float tos(time=1980, j=300, i=360);
:standard_name = "sea_surface_tem
:long_name = "Sea Surface Tempera
:comment = "Temperature of upper
:units = "degC";
:cell_methods = "area: mean where
: areacello
T18:45:39Z
0f; // floa
// float
de longitud
U, 360U; //
```

Create Plot

More than one type of plot can be created from the variable 'tos'. What type would you like to create?

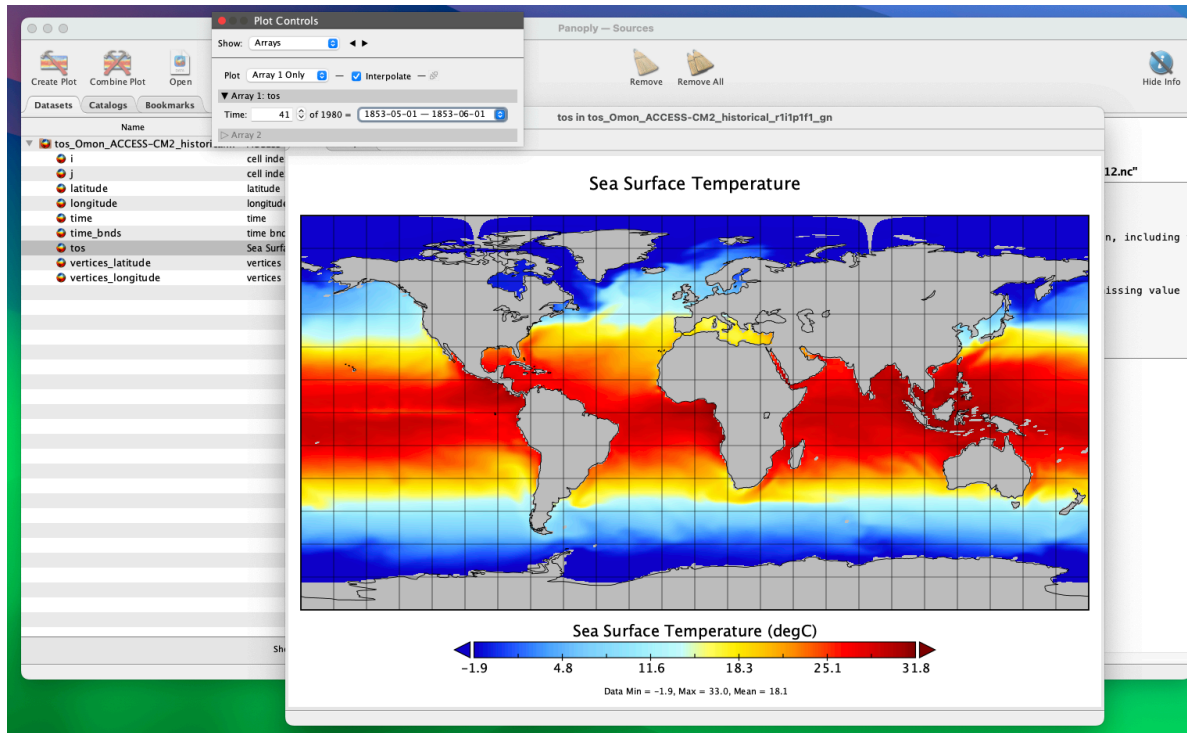
☒ Georeferenced Longitude-Latitude color contour plot

☐ Georeferenced Zonal Average line plot

☐ Color contour plot using i for X axis and j for Y axis

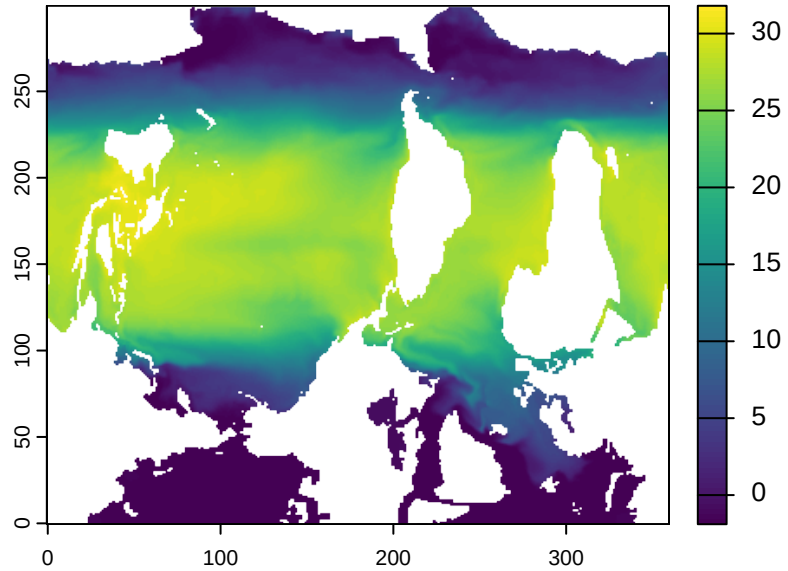
☐ Line plot using i for the horizontal axis

Cancel Create



3.1.2 R

```
rr <- rast(paste0(pth, cmip_pth, "/tos_Omon_ACCESS-CM2_historical_r1i1p1f1_gn_185001-201412."),
rr <- rr[[1]] #the first layer
plot(rr)
```



3.2 Remapping

🔥 Caution

You can also use **first order conservative remapping** if bilinear interpolation fails (which may occur if the input grid is ‘unstructured’ as opposed to ‘spherical’).

rr

! We can also check more detailed metadata contained in the ESM file using `ncdf4::nc_open()`

This gives you the same information after reading a file into Panoply.

```
nc <- nc_open(paste0(pth, cmip_pth, "/tos_Omon_ACCESS-CM2_historical_r1i1p1f1_gn_185001-20140101"),  
nc
```

File /Users/admin/Documents/GitHub/esmRworkshop_notes/__data/cmip6_raw/tos_Omon_ACCESS-CM2_historical_r1i1p1f1_gn_185001-20140101.nc

```
6 variables (excluding dimension variables):  
  double time_bnds[bnds,time]    (Chunking: [2,1]) (Compression: level 1)  
  double latitude[i,j]          (Chunking: [360,300]) (Compression: level 1)  
    standard_name: latitude  
    long_name: latitude  
    units: degrees_north  
    missing_value: 1e+20  
    _FillValue: 1e+20  
    bounds: vertices_latitude  
  double longitude[i,j]         (Chunking: [360,300]) (Compression: level 1)  
    standard_name: longitude  
    long_name: longitude  
    units: degrees_east  
    missing_value: 1e+20  
    _FillValue: 1e+20  
    bounds: vertices_longitude  
  double vertices_latitude[vertices,i,j] (Chunking: [2,360,300]) (Compression: level 1)  
    units: degrees_north  
    missing_value: 1e+20  
    _FillValue: 1e+20  
  double vertices_longitude[vertices,i,j] (Chunking: [2,360,300]) (Compression: level 1)  
    units: degrees_east  
    missing_value: 1e+20
```

```

        _FillValue: 1e+20
float tos[i,j,time]    (Chunking: [360,300,1]) (Compression: level 1)
    standard_name: sea_surface_temperature
    long_name: Sea Surface Temperature
    comment: Temperature of upper boundary of the liquid ocean, including temperature
    units: degC
    cell_methods: area: mean where sea time: mean
    cell_measures: area: areacello
    history: 2019-11-08T18:45:39Z altered by CMOR: replaced missing value flag (-1e+20)
    missing_value: 1.00000002004088e+20
    _FillValue: 1.00000002004088e+20
    coordinates: latitude longitude

5 dimensions:
    time Size:1980    *** is unlimited ***
        bounds: time_bnds
        units: days since 1850-01-01
        calendar: proleptic_gregorian
        axis: T
        long_name: time
        standard_name: time
    j Size:300
        units: 1
        long_name: cell index along second dimension
    i Size:360
        units: 1
        long_name: cell index along first dimension
    bnds Size:2 (no dimvar)
    vertices Size:4 (no dimvar)

47 global attributes:
    Conventions: CF-1.7 CMIP-6.2
    activity_id: CMIP
    branch_method: standard
    branch_time_in_child: 0
    branch_time_in_parent: 0
    creation_date: 2019-11-08T18:45:44Z
    data_specs_version: 01.00.30
    experiment: all-forcing simulation of the recent past
    experiment_id: historical
    external_variables: areacello

```



```

forcing_index: 1
frequency: mon
further_info_url: https://furtherinfo.es-doc.org/CMIP6.CSIRO-ARCCSS.ACCESS-CM2.hist
grid: native atmosphere N96 grid (144x192 latxlon)
grid_label: gn
history: 2019-11-08T18:45:44Z ; CMOR rewrote data to be consistent with CMIP6, CF-
initialization_index: 1
institution: CSIRO (Commonwealth Scientific and Industrial Research Organisation, A
institution_id: CSIRO-ARCCSS
mip_era: CMIP6
nominal_resolution: 250 km
notes: Exp: CM2-historical; Local ID: bj594; Variable: tos (['sst'])
parent_activity_id: CMIP
parent_experiment_id: piControl
parent_mip_era: CMIP6
parent_source_id: ACCESS-CM2
parent_time_units: days since 0950-01-01
parent_variant_label: r1i1p1f1
physics_index: 1
product: model-output
realization_index: 1
realm: ocean
run_variant: forcing: GHG, Oz, SA, Sl, Vl, BC, OC, (GHG = CO2, N2O, CH4, CFC11, CFC
source: ACCESS-CM2 (2019):
aerosol: UKCA-GLOMAP-mode
atmos: MetUM-HadGEM3-GA7.1 (N96; 192 x 144 longitude/latitude; 85 levels; top level 85 km)
atmosChem: none
land: CABLE2.5
landIce: none
ocean: ACCESS-OM2 (GFDL-MOM5, tripolar primarily 1deg; 360 x 300 longitude/latitude; 50 lev
ocnBgchem: none
seaIce: CICE5.1.2 (same grid as ocean)
    source_id: ACCESS-CM2
    source_type: AOGCM
    sub_experiment: none
    sub_experiment_id: none
    table_id: Omon
    table_info: Creation Date:(30 April 2019) MD5:e14f55f257cceafb2523e41244962371
    title: ACCESS-CM2 output prepared for CMIP6
    variable_id: tos
    variant_label: r1i1p1f1

```

```
version: v20191108
cmor_version: 3.4.0
tracking_id: hdl:21.14100/0bcaaa74-aedb-4d45-a5e5-cb3ab467f2b5
license: CMIP6 model data produced by CSIRO is licensed under a Creative Commons Attribution 4.0 International License
```

3.2.1 A (brief) intro to CDO

Common CDO operators

```
cdo -yearmean calculates the annual mean of a monthly data input netCDF file
cdo -yearmin calculates the annual min of a monthly data input netCDF file
cdo -yearmax calculates the annual max of a monthly data input netCDF file
cdo -ensmean calculates the ensemble mean of several netCDF files. If your input files
are different models, this function will estimate a mean of all those models
cdo -vertmean calculates the vertical mean for netCDF with olevel (i.e., depth)
cdo -mergetime merge all the netCDF files in your directory
```

3.2.2 Remapping with CDO and R

```
remap_n_crop_temp <- function(nc_file,
                             cell_res = 0.25,
                             infold = paste0(pth, cmip_pth),
                             outfold = paste0(pth, cmip_pth_proc),
                             xmin = -126, xmax = -115, ymin = 32, ymax = 43) {

  system(paste0("cdo -L -sellonlatbox,", xmin, ",", xmax, ",", ymin, ",", ymax,
                " -remapbil,r", 360*(1/cell_res), "x", 180*(1/cell_res),
                " -select,name=tos ", infold, "/", nc_file, " ", outfold, "/", nc_file))

}

fileys <- list.files(paste0(pth, cmip_pth)) #list file names of downloaded ESMs

w <- 14 #number of workers
plan(multisession, workers = w) # Change to multi-threaded processing
tic(); future_walk(fileys, remap_n_crop_temp); toc() #Run the function in parallel (takes 8
plan(sequential) # Return to single threaded processing
```

3.2.3 Function explainer

- •
•
•
•

•
•
•
•
•

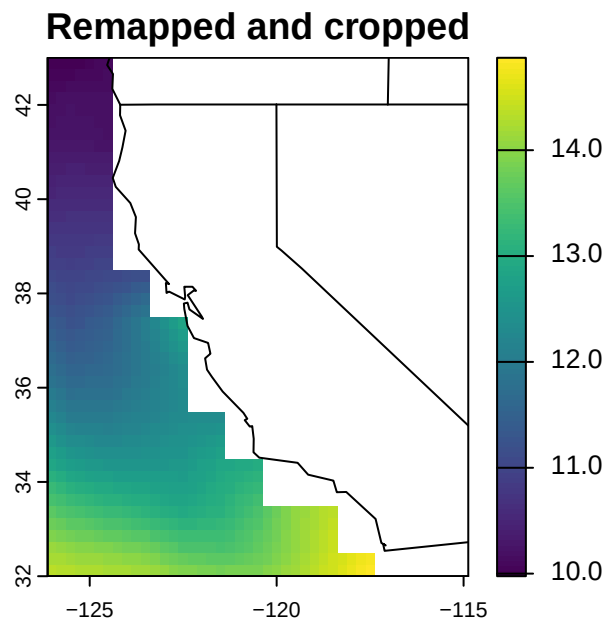
-
-
-
-

! Important

CDO requires syntax (i.e, spaces and commas) to be precise. If your code doesn't run, first check that you have them in the right places.

3.3 Check that it worked

```
# Read in one of the processed output files
rr <- rast(paste0(pth, cmip_pth_proc, "/tos_Omon_ACCESS-CM2_historical_r1i1p1f1_gn_185001-201401-201412"),
           bylayer = TRUE)
rr <- rr[[1]] #first layer
plot(rr, main = "Remapped and cropped")
maps::map("state", add = T) #add US state boundaries
```



```
rr #display metadata
```

4 OISST

i Learning objective:

4. Observations (theory and live coding)

- Identify suitable observational datasets (e.g., OISST) for bias correction.
- Regrid and prepare observational datasets for comparison with ESM outputs.

OISST V2.1

i Oceanography terminology: ‘observations’

Whether data are referred to as **observations** depends on who you talk to. For the purposes of our workshop, and as marine ecologists, we refer to OISST as our ‘observed’ data set, because it represents ‘real-life’ SST. However, sea-going physical oceanographers might not refer to OISST as observations, because the product represents a series of observations obtained from different platforms (i.e., ARGO, moorings, satellites, ship-borne sensors etc.) that have been ‘modeled’ or ‘interpolated’ into a gridded, cloud-free product. Technically, OISST represents a modeled product *based* on observations. Rather, they would refer to observations as the raw, unprocessed data coming from sensors themselves. Just something to keep in mind when talking to people from different disciplines.

4.1 Download OISST

[Here](#)

Download and Plot Data

Data Help

Search Dataset Variables (start with * to match anywhere)

Clear

+ / -

	Variable	Statistic	Level	TimeScale	Options
	Temperature	Mean			
1	Sea Surface Temperature	Long Term Mean	Surface	Monthly	 
45	Sea Surface Temperature	Mean	Surface	Daily	  
2	Sea Surface Temperature	Monthly Mean	Surface	null	  
1	Sea Surface Temperature	Monthly Mean	Surface	Monthly	  
	/Datasets/noaa.oisst.v2.highres/sst.mon.mean.nc				  
1	Sea Surface Temperature	Weekly Mean	Surface	Weekly	   Catalog

Thredds Catalog:  Download:  Plot/Subset: 



THREDDS Data Server

Catalog: <https://psl.noaa.gov/thredds/catalog/Datasets/noaa.oisst.v2.highres/catalog.html>

Dataset: [Datasets/noaa.oisst.v2.highres/sst.mon.mean.nc](#)

Data Type: Grid

Data Size: 2177.35 Mbytes

ID: [Datasets/noaa.oisst.v2.highres/sst.mon.mean.nc](#)

Access Viewers

Service	Type	Description
OpenDAP	Data Access	Access dataset through OPeNDAP using the DAP2 protocol.
HTTPServer	Data Access	HTTP file download.
WCS	Data Access	Supports access to geospatial data as 'coverages'.
WMS	Data Access	Supports access to georegistered map images from geoscience datasets.
NetcdfSubset	Data Access	A web service for subsetting CDM scientific grid datasets.
ISO	Metadata	Provide ISO 19115 metadata representation of a dataset's structure and metadata.
NCML	Metadata	Provide NCML representation of a dataset.
UDDC	Metadata	An evaluation of how well the metadata contained in the dataset conforms to the NetC

Dates

-
-
-

NetCDF Subset Service for Grids

([Switch to Grids As Points](#))

Dataset: /thredds/ncss/grid/Datasets/noaa.oisst.v2.highres/sst.mon.mean.nc

([Dataset Description](#))

Variables:

With coordinate axes: **time lat lon** ▼

☒ *Monthly Mean of Sea Surface Temperature*



Horizontal subset:

Lat/lon box

BOUNDING BOX (IN DECIMAL DEGREES)

North

90.000

West -180.000 East 180.000

-90.000

South

Reset to full extension

Horizontal stride: 1

Time subset:

Time range **Single time**

Start: 1995-01-01T00:00:00Z

End: 2014-12-01T00:00:00Z

Stride:

Reset to full extension

Vertical subset:

Single level **Vertical stride**

Level:

Output format:

Format: netcdf4-classic ▼

CF compliance:

☒ Add 2D Lat/Lon to file

<https://psl.noaa.gov/thredds/ncss/grid/Datasets/noaa.oisst.v2.highres/sst.mon.mean.nc?var=sst>

4.2 Preprocess OISST

```
oisst_mr <- function(infile = paste0(pth, oisst_pth),
                      outfile = paste0(pth, oisst_pth_proc),
                      xmin = -126, xmax = -116, ymin = 32, ymax = 43,
                      cell_res = 0.25) {

  # Select SST, crop and remap
  oisst_regrid <- paste0("cdo -L -select,name=sst ",
                        "-sellonlatbox,", xmin, ",", xmax, ",", ymin, ",", ymax,
                        " -remapbil,r", 360*(1/cell_res), "x", 180*(1/cell_res),
                        " ", infile, "/sst.mon.mean.nc.nc4",
                        " ", outfile, "/sst.mon.mean_remap.nc")
  system(oisst_regrid)
}

oisst_mr() #run with default function arguments
```

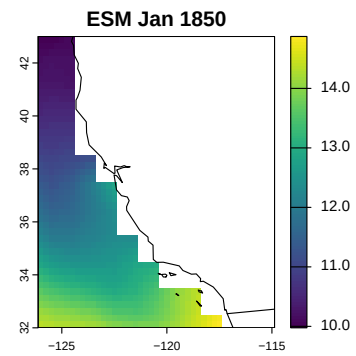
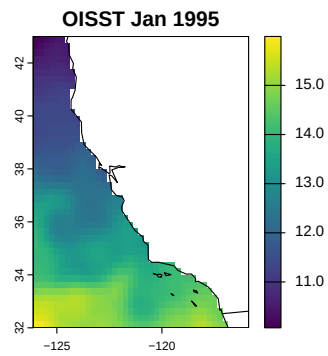
```
list.files(paste0(pth, "/__data/oisst_processed"))
```

4.3 Visualise

```

oisst <- terra::rast(paste0(pth, oisst_pth_proc, "/sst.mon.mean_remap.nc"))[[1]]
esm <- terra::rast(paste0(pth, cmip_pth_proc, "/tos_0mon_ACCESS-CM2_historical_r1i1p1f1_gn_1"))
terra::plot(oisst, main = "OISST Jan 1995")
maps::map("world", add = T)
terra::plot(esm, main = "ESM Jan 1850")
maps::map("world", add = T)

```



5 Bias correction & downscaling

 Learning objective:

5. Bias correction (theory and live coding)

- Explain the purpose of bias correction and the delta method.
- Apply the delta method to generate bias-corrected climate projections in R.

 Tip

In this chapter, we are coding in R, but R is calling CDO commands through the terminal. See the Resources page for more information on CDO and how the tool works.

 Important

Note - a lot of physical oceanographers caution against interpolating or bias-correcting ESMs to a (much) higher resolution, as results can often be misleading. Today, we'll show you how to downscale an ESM from 1° to 0.25° and perform bias-correction; but always best to chat to an oceanographer first, when working with your own study system.

! Defining climatologies

We often use 30 years to define a climatology in marine ecology, but you can shorten or lengthen your climatology based on what makes sense for your question and study system. We used 20 years today to save time.

Further, in your own work, you may want to calculate monthly or seasonal climatologies, as opposed to a broad multi-decadal climatology (because this will smooth out monthly or seasonal variability). This will all depend on your study system and question, but something to consider when bias correcting ESM output.

5.1 Create OISST climatology (1995-2014)

! Defining ‘climatologies’

Here, for simplicity, we’ve defined our ‘climatology’ as the SST averaged over the 20 year period between 1995-2014. However, you may want to consider calculating monthly climatologies (i.e., average SST over Jan, Feb, March etc. across the 20 year period), or even seasonal climatologies, so that you can capture and remove this sort of variability during the step of bias correction, for example. We highly recommend consulting with a physical oceanographer if doing bias correction and have questions about defining climatologies and temporal modes of variability.

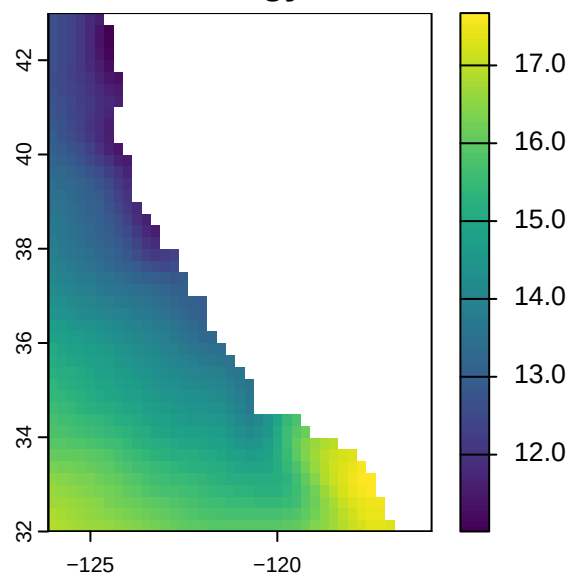
```
input_file <- paste0(pth, "/_data/oisst_processed/sst.mon.mean_remap.nc")
output_file <- paste0(pth, bc_pth, "/_2_OISST_climatology.nc")

# Calculate the mean in each grid cell (across all years in the time period)
cdo_code <- paste0("cdo timmean ", input_file, " ", output_file )
system(cdo_code)
```

```
r <- terra::rast(paste0(pth, bc_pth, "/_2_OISST_climatology.nc"))
r #ignore 'time' field
```

```
plot(r, main = "OISST climatology 1995-2014") #smoothed over
```

OISST climatology 1995-2014



5.2 Create ESM historical climatology (1995-2014)

```
histclim <- function(model) {  
  
  ### HISTORICAL  
  # This creates a climatology/baseline 1995-2014 file using the ESM's historical run  
  # Create baseline file 1993-2014 and mean
```



```

filey <- list.files(paste0(pth, cmip_pth_proc),
                    pattern = "historical", full.names = T)
filey <- filey[grep(model, filey)]
rr <- rast(filey)
#time(rr) <- as.Date(time(rr))
rr <- rr[[time(rr) > as.Date("1995-01-01")]]
mean_rr <- mean(rr)
terra::writeCDF(mean_rr,
                paste0(pth, bc_esm_pth, "/_1_climatology/tos_mo_", model,
                      "_1995-2014_clim_historical.nc"),
                overwrite = T)

}

histclim("ACCESS-CM2")
histclim("IPSL-CM6A-LR")

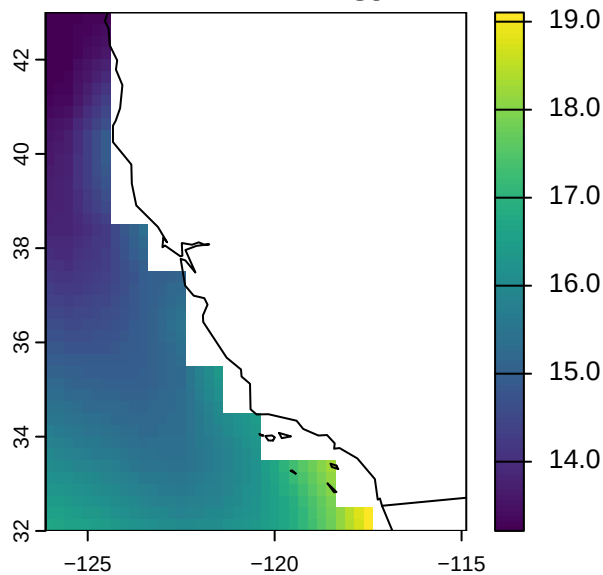
```

```

clim <- terra::rast(paste0(pth, bc_esm_pth, "/_1_climatology/tos_mo_ACCESS-CM2_1995-2014_clim_historical.nc"))
plot(clim, main = "ACCESS-CM2 climatology 1995-2014"); maps::map("world", add = T)

```

ACCESS-CM2 climatology 1995-2014



5.3 Create anomalies for ESM projections (2015-2100)

```
anom_esm <- function(model) {

  mean_rr <- rast(paste0(pth, bc_esm_pth, "/_1_climatology/tos_mo_", model,
    "_1995-2014_clim_historical.nc"))

  ### SSPs
  # Then subtract mean climatology from 2015-2100 for both EMSs, creating anomalies
  filey <- list.files(paste0(pth, cmip_pth_proc),
    pattern = "ssp", full.names = T)
  filey <- filey[grep(model, filey)]

  for (i in 1:length(filey)) {

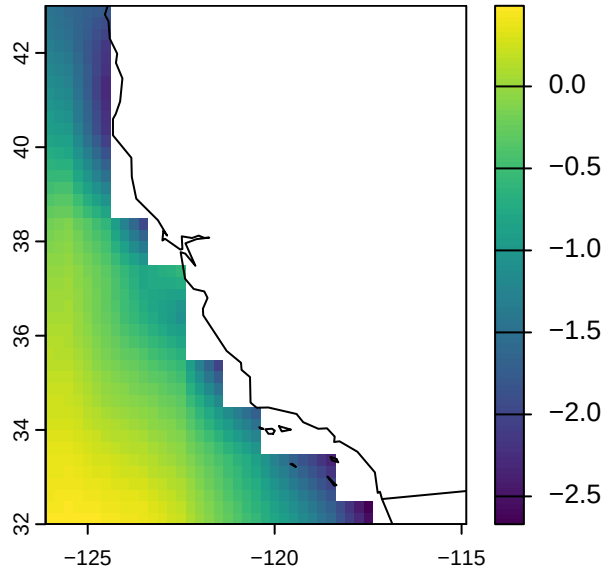
    #isolate SSP from string
    ssp_string <- strsplit(filey[i], model, "_")[[1]][2]
    ssp <- strsplit(ssp_string, "_r1i1p1f1")[[1]][1]

    #import raster and calc anomalies
    scen <- rast(filey[i])
    anom <- scen - mean_rr
    writeCDF(anom, paste0(pth, bc_esm_pth, "/_2_anomalies/tos_mo_",
      model, "_2015-2100_anom", ssp, ".nc"),
      overwrite = T)
  }
}

# Run function
tic(); anom_esm("ACCESS-CM2"); toc() #Jessie: 0.689 seconds
anom_esm("IPSL-CM6A-LR")
```

```
rr <- terra::rast(paste0(pth, bc_pth, bc_pth_anom, "/tos_mo_ACCESS-CM2_2015-2100_anom_ssp245
plot(rr[[1]], main = "ACCESS-CM2 anomalies SSP245 Jan 2015"); maps::map("world", add = T)
```

ACCESS-CM2 anomalies SSP245 Jan 2015

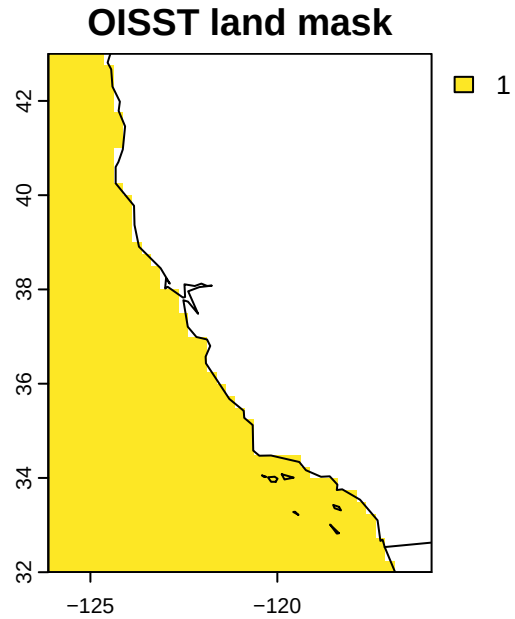


5.4 Create a land mask with OISST

```
infile <- paste0(pth, bc_pth, "/_2_OISST_climatology.nc")
outfile <- paste0(pth, bc_pth, "/_3_OISST_mask.nc")

# make a mask saying "NA:land, 1:ocean"
system(paste0("cdo -expr,'sst = ((sst>-2)) ? 1.0 : sst/0.0' ", infile, " ", outfile))

#Plotting the mask results in...
r <- rast(outfile)
plot(r, main = "OISST land mask"); values(r) %>% range(na.rm=T); maps::map("world", add = T)
```



5.5 Interpolate ESM anomalies to OISST grid

```
input_folder <- paste0(pth, bc_pth, "/", bc_pth_anom)
output_folder <- paste0(pth, bc_pth, "/", bc_pth_anom_rm)
msk <- paste0(pth, bc_pth, "/_3_OISST_mask.nc")

# Function to do the re-gridding -----

do_regrid <- function(f) {

  output_file <- basename(f) %>% # Get input file name
    gsub(".nc", "_remapped.nc", .) %>% # Add "_remapped" to end of file name
```

```

    paste0(output_folder, "/", .)

# Step 1 - set grid via nco
cdo_code <- paste0('ncatted -O -a units,longitude,c,c,"degrees_east" -a units,latitude,c,c,
                  f, " ", paste0(output_folder, "/tmp1.nc"))
system(cdo_code)

# Get variable name
rr <- rast(paste0(output_folder, "/tmp1.nc"))
vary <- varnames(rr)

# Step 2 - remove lat/lon attributes
cdo_code2 <- paste0('ncatted -O -a grid_mapping,', vary, ',d,, ',
                  paste0(output_folder, "/tmp1.nc"), ' ',
                  paste0(output_folder, "/tmp2.nc"))
system(cdo_code2)

# Step 3 - remap ESM anomaly with OISST mask using bilinear interpolation
# Set missing vals to nearest neighbour
cdo_code3 <- paste0("cdo -s -L -f nc4 -z zip ", # Zip the file up
                  "-setmisstonn ", # Set missing values to nearest neighbor value
                  "-remapbil,", msk, " ", # Remap with OISST mask sing bilinear interpolation
                  paste0(output_folder, "/tmp2.nc"), " ", #input file
                  paste0(output_folder, "/tmp3.nc")) # output file
system(cdo_code3)

# Step 4 - remove mask
cdo_code <- paste0("cdo -s -L -f nc4 -z zip ", # Zip the file up
                  "-mul ", msk, " ", # Multiply the result of the lines, below by the mask
                  paste0(output_folder, "/tmp3.nc"),
                  " ", output_file) # Mask the remapped anomaly file, and save as output.
system(cdo_code)

# Step 5 -remove temporary files
system(paste0("rm ", output_folder, "/tmp1.nc", " ",
             output_folder, "/tmp2.nc ",
             output_folder, "/tmp3.nc"))
}

# Run function -----
files <- list.files(input_folder, pattern = "anom", full.names = TRUE) # Files to process

```

```

plan(multisession, workers = 14) # Setting up to run in parallel, change workers to suit you
tic(); frrrr::future_walk(files, do_regrid); toc()
plan(sequential) # Go back to sequential processing

# If having trouble with using future_walk(), you can try the for loop version.
# However, this takes longer, since it iterates over each file one at a time!
# for (f in files) {
#   do_regrid(f)
#   print(f)
# }

```

! There's a lot going on in this function. Let's break it down...

5.5.1 Create output_file name

Takes the input file (e.g., `tos_mo_ACCESS-CM2_2015-2100_anom_ssp245.nc`), and adds `_remapped` to the end of the file name.

```

output_file <- basename(f) %>% # Get input file name
  gsub(".nc", "_remapped.nc", .) %>% # Add "_remapped" to end of file name
  paste0(output_folder, "/", .)

output_file
#[1] "/Users/admin/Documents/GitHub/BMLworkshop/__data/bias_correct/esm/_3_anomalies_remap

```

5.5.2 Set grid via NCO

This takes the file created above, and uses the `ncatted` command from NCO to modify the input file's `longitude` and `latitude` variables' attribute units, setting them to `degrees_east` and `degrees_north`, respectively. The output is saved as `tmp1.nc`. We then get the variable name from the file, which will be the original name of the file, e.g., `tos_mo_ACCESS-CM2_2015-2100_anom_ssp245`.

```

cdo_code <- paste0('ncatted -O -a units,longitude,c,c,"degrees_east" -a units,latitude,c,c,
  f, " ", paste0(output_folder, "/tmp1.nc"))
system(cdo_code)

rr <- rast(paste0(output_folder, "/tmp1.nc"))
vary <- varnames(rr)

```

5.5.3 Remove lon/lat attributes

This code removes the `grid_mapping` attribute from the variable(s) specified in `vary`, and save the output as `tmp2.nc`.

```
cdo_code2 <- paste0('ncatted -O -a grid_mapping,', vary, ',d,, ',  
                    paste0(output_folder, "/tmp1.nc"), ' ',  
                    paste0(output_folder, "/tmp2.nc"))  
system(cdo_code2)
```

5.5.4 Remapping

Here, we remap the ESM anomalies with the OISST mask `msk` using bilinear interpolation, and set missing values to nearest neighbor. The output is saved as `tmp3.nc`.

```
cdo_code3 <- paste0("cdo -s -L -f nc4 -z zip ", # Zip the file up  
                    "-setmisstonn ", # Set missing values to nearest neighbor value  
                    "-remapbil,", msk, " ", # Remap with OISST mask using bilinear interpolation  
                    paste0(output_folder, "/tmp2.nc"), " ", #input file  
                    paste0(output_folder, "/tmp3.nc")) # output file  
system(cdo_code3)
```

5.5.5 Remove the OISST mask

This code uses the OISST mask to turn land values back to NA.

```
cdo_code <- paste0("cdo -s -L -f nc4 -z zip ", # Zip the file up  
                  "-mul ", msk, " ", # Multiply contents of file with the mask to make land values NA  
                  paste0(output_folder, "/tmp3.nc"),  
                  " ", output_file) # Mask the remapped anomaly file, and save as output file  
system(cdo_code)
```

5.5.6 Remove temporary files

We use the `rm` command (i.e., a base LINUX function) to remove the three temporary files created previously.

```
system(paste0("rm ", output_folder, "/tmp1.nc", " ",  
              output_folder, "/tmp2.nc",  
              output_folder, "/tmp3.nc"))
```

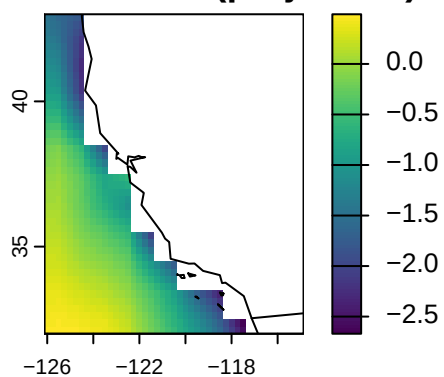
5.5.7 Run the function

Here, we use `list.files()` to list all files in our `input_folder` with the pattern `anom` in the file name. We then set up our session to run in parallel, where I (Jessie) have set to 14 workers (you will need to change this to suit your machine). We then run the `do_regrid()` function in parallel using `future_walk`. Once the function has finished running, we set our session back to normal (i.e., sequential) processing.

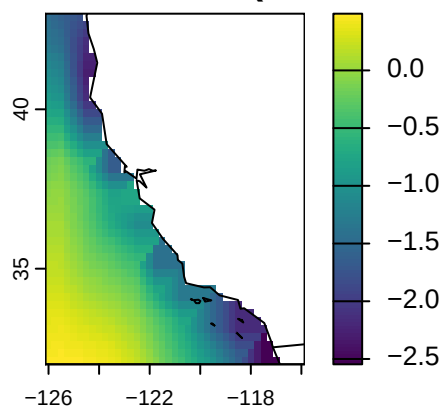
```
files <- list.files(input_folder, pattern = "anom", full.names = TRUE) # Files to process
plan(multisession, workers = 14) # Setting up to run in parallel, change workers to suit y
tic(); frrrr::future_walk(files, do_regrid); toc() #Jessie: 5.7 seconds
plan(sequential) # Go back to sequential processing
```

```
par(mfrow=c(1,2))
rast(paste0(pth, bc_pth, bc_pth_anom, "/tos_mo_ACCESS-CM2_2015-2100_anom_ssp245.nc"))[[1]] %
  plot(main = "ESM anomalies (proj - hist)")
maps::map("world", add =T)
rast(paste0(pth, bc_pth, bc_pth_anom_rm, "/tos_mo_ACCESS-CM2_2015-2100_anom_ssp245_remapped
  plot(main = "Remapped anomalies (OISST mask)")
maps::map("world", add =T)
```


ISM anomalies (proj - hist)



Open anomalies (OISST mask)



i Troubleshooting CDO gridding errors

If you ever receive errors when using CDO relating to gridding, projections or remapping, it's a good idea to check the grid structure of the file using `griddes`, and ensure that the files you are manipulating/trying to create have the same grid structure as your input file.

```
system(paste0("cdo griddes ", inputfile))
```

5.6 Bias correction

```
input_folder <- paste0(pth, bc_esm_pth)
obs <- paste0(pth, bc_pth, "/_2_OISST_climatology.nc")
```

```

# Function -----

do_add_clim <- function(f) {

  output_file <- basename(f) %>%
    gsub("_anom_", "_bc_", .) %>% # Replace anom code in file name with a code for bias corr
    paste0(input_folder, "/_4_bias_corrected/", .) # Include the path

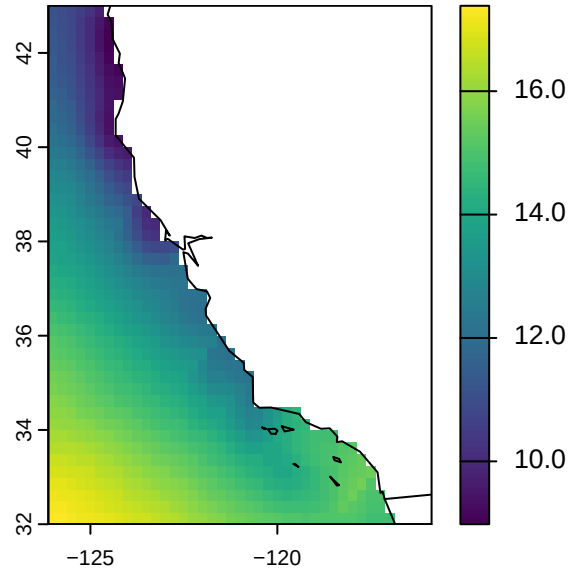
  cdo_code <- paste0("cdo -s -L -f nc4 -z zip ", # Zip the file up
    "-add ", f, " ", # To the remapped regrided anomalies, add...
    obs, " ", output_file) # The observed climatology (map of means)
  system(cdo_code)
}

# Run function -----

files <- list.files(paste0(input_folder, "/_3_anomalies_remapped"),
  pattern = "remapped", full.names = TRUE) # The files we want to process
tic(); walk(files, do_add_clim); toc()

```

Bias corrected w/OISST clim



```
par(mfrow=c(2,2))

input_folder <- paste0(pth, "/_data/bias_correct/esm")

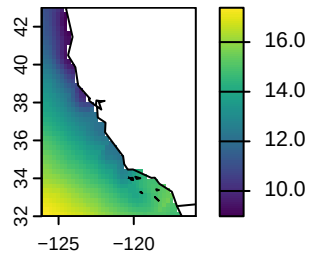
rr <- rast(paste0(input_folder, "/_4_bias_corrected/tos_mo_ACCESS-CM2_2015-2100_bc_ssp245_remap.nc"))
plot(rr, main = "Bias corrected w/OISST clim")
maps::map("world", add = T)

tt <- rast(paste0(pth, cmip_pth_proc, "/tos_Omon_ACCESS-CM2_ssp245_r1i1p1f1_gn_201501-210012.nc"))
plot(tt, main = "Raw ESM")
maps::map("world", add = T)

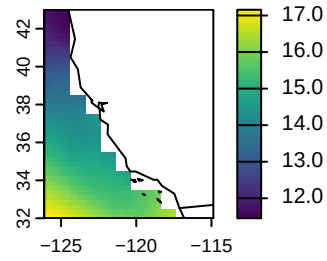
rr <- rast(paste0(pth, bc_esm_pth, "/_3_anomalies_remapped/tos_mo_ACCESS-CM2_2015-2100_anom_ssp245_remap.nc"))
plot(rr, col = viridis::magma(255), main = "Remapped anomalies w/OISST")
maps::map("world", add = T)

rr <- rast(paste0(pth, bc_esm_pth, "/_2_anomalies/tos_mo_ACCESS-CM2_2015-2100_anom_ssp245.nc"))
plot(rr, col = viridis::magma(255), main = "Anomalies Raw ESM")
maps::map("world", add = T)
```

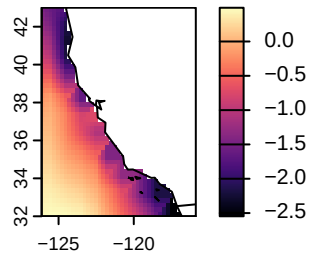
Bias corrected w/OISST clim



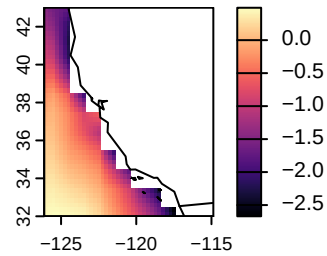
Raw ESM



Remapped anomalies w/OISST



Anomalies Raw ESM



5.7 Rinse and repeat

```
list.files(paste0(pth, "/__data/bias_correct/esm/_4_bias_corrected"),  
          pattern = "1995-2014")
```

6 Evaluate accuracy of historical ESM projections

i Learning objective:

6. Evaluating ESM accuracy (live coding)

- Evaluate ESM performance during the observational period using quantitative metrics and visualizations (e.g., Taylor diagrams).
- Interpret model performance to inform model selection for ecological applications.

6.1 Brief explainer

-
-
-

6.2 Create dataframes of baseline climatologies

```
source(paste0(pth, "/_scripts/helpers.R"))
```

```
library(terra)
library(tidyverse)
library(plotrix)

oisst <- rast(paste0(pth, bc_pth, "/_2_OISST_climatology.nc"))
access_bc <- rast(paste0(pth, bc_pth, bc_pth_bc, "/tos_mo_ACCESS-CM2_1995-2014_bc_historical_"))
ipsl_bc <- rast(paste0(pth, bc_pth, bc_pth_bc, "/tos_mo_IPSL-CM6A-LR_1995-2014_bc_historical_"))
access_raw <- rast(paste0(pth, cmip_pth_proc, "/tos_Omon_ACCESS-CM2_historical_r1i1p1f1_gn_1"))
access_raw <- access_raw[[time(access_raw) > "1994-01-01"]]
ipsl_raw <- rast(paste0(pth, cmip_pth_proc, "/tos_Omon_IPSL-CM6A-LR_historical_r1i1p1f1_gn_1"))
ipsl_raw <- ipsl_raw[[time(ipsl_raw) > "1994-01-01"]]

# Convert rasters to dataframes -----

# Calculate mean of each field (i.e., baseline/climatology)
oisst_df <- oisst %>% mean %>% as.data.frame(xy = T)
access_df <- access_raw %>% mean %>% as.data.frame(xy = T)
access_bc_df <- access_bc %>% mean %>% as.data.frame(xy = T)
ipsl_df <- ipsl_raw %>% mean %>% as.data.frame(xy = T)
ipsl_bc_df <- ipsl_bc %>% mean %>% as.data.frame(xy = T)

# Merge and fix names -----

alldata <- inner_join(oisst_df, access_df, by = c("x", "y"))
names(alldata) <- c("x", "y", "oisst_mean", "access_mean")
alldata <- inner_join(alldata, access_bc_df, by = c("x", "y"))
names(alldata) <- c("x", "y", "oisst_mean", "access_mean", "access_bc_mean")

alldata2 <- inner_join(oisst_df, ipsl_df, by = c("x", "y"))
names(alldata2) <- c("x", "y", "oisst_mean", "ipsl_mean")
alldata2 <- inner_join(alldata2, ipsl_bc_df, by = c("x", "y"))
names(alldata2) <- c("x", "y", "oisst_mean", "ipsl_mean", "ipsl_bc_mean")
```

```

alldata <- inner_join(alldata, alldata2)

alldata <- alldata %>% #ensemble mean
  mutate(ens_bc_mean = (access_bc_mean + ipsl_bc_mean) / 2)

alldata %>% head

```

6.3 Construct Taylor Diagram

```

taylor.diagram(alldata$oisst_mean,
               alldata$oisst_mean,
               ref.sd = T, #display arc of ref. std. dev. (i.e., 1)
               normalize=TRUE, #normalize models so ref has SD of 1

```

```

sd.arcs=TRUE, #display arcs along SD axes
pcex = 4,
pch = 19,
col = "red",
xlab = "Standard deviation (normalised)",
pos.cor = T, #show correlation (y-axis) from 0-1
gamma.col = "blue", #RMSE arcs
main="OISST vs. CMIP6 ESM tos (SST) 1995-2014")

# Add ESM points
taylor.diagram(alldata$oisst_mean,
               alldata$access_mean,
               add=TRUE, normalize=TRUE,
               pcex=3, pch=17, col= "purple")

taylor.diagram(alldata$oisst_mean,
               alldata$ips1_mean,
               add=TRUE, normalize=TRUE,
               pcex=3, pch=17, col= "forestgreen")

# Add bias-corrected point
taylor.diagram(alldata$oisst_mean,
               alldata$access_bc_mean,
               add=TRUE, normalize=TRUE,
               pcex=2, pch=17, col= "black")

# Legend -----

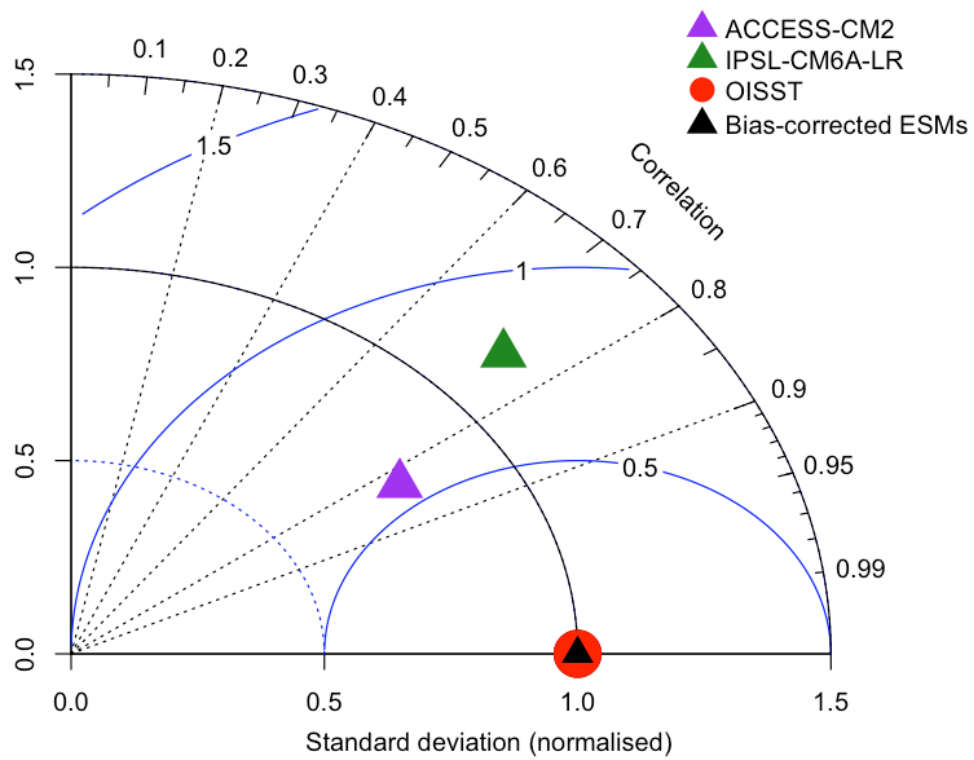
legend(1.2, 1.7, cex=1, pt.cex=2, pch=17,
       legend=c("ACCESS-CM2", "IPSL-CM6A-LR"),
       col=c("purple", "forestgreen"),
       bty = "n")

legend(1.2, 1.54, cex=1, pt.cex=2, pch=19,
       legend=c("OISST"),
       col= 'red',
       bty = "n")

legend(1.2, 1.45, cex=1, pt.cex=2, pch=17,
       legend=c("Bias-corrected ESMs"),
       col= 'black',
       bty = "n")

```


OISST vs. CMIP6 ESM tos (SST) 1994-2014



7 Making projections

i Learning objective:

7. Making projections (live coding)

- Generate future sea-surface temperature projections from the two CMIP6 models and two climate scenarios.
- Visualize projected changes through time series plots and spatial maps.

7.1 Time series

7.1.1 Create yearly averages of temperature

```
models = c("ACCESS-CM2|IPSL-CM6A-LR")
models2 = c("ACCESS-CM2", "IPSL-CM6A-LR")

indir_proj <- paste0(pth, "/__data/bias_correct/esm/_4_bias_corrected")
outdir <- "__data/timeseries"

# Projections -----
```

```

timeseries_ssp <- function(ssp) {

  allfiles_proj <- list.files(indir_proj, pattern = models, full.names = T)
  allfiles_proj2 <- allfiles_proj[grepl(ssp, allfiles_proj)]
  rr <- rast(allfiles_proj2)
  rr <- rr[[time(rr) < "2100-12-31"]] #Only data until 2100
  dateys <- time(rr) %>% unique
  years <- lubridate::year(dateys) %>% unique #get years

  emplist <- list()
  emplist_allmodels <- list()

  ### Ensemble mean
  for (i in 1:length(years)) {

    alldates <- dateys[grepl(years[i], dateys)]
    rasty <- rr[[time(rr) == alldates]]
    meanrast <- mean(rasty) # Average across the year
    meanval <- values(meanrast) %>% mean(na.rm=T) #Get average temp for California
    forlist <- data.frame(date = years[i], value = meanval)
    emplist[[i]] <- forlist
    print(paste0("ens_", years[i]))
  }

  ### Individual models
  for (j in 1:length(models2)) {
    emplist_indmodel <- list()

    for (h in 1:(length(years)-1)) {
      modelrast <- rast(allfiles_proj2[grepl(models2[j], allfiles_proj2)])

      dateys <- time(modelrast) %>% unique
      alldates <- dateys[grepl(years[h], dateys)]
      modelrast <- modelrast[[time(modelrast) == alldates]]
      meanrast <- mean(modelrast)
      meanval <- values(meanrast) %>% mean(na.rm=T)
      forlist <- data.frame(model = models2[j], date = years[h], value = meanval)
      emplist_indmodel[[h]] <- forlist
    }

    toadd <- do.call(rbind, emplist_indmodel)
    emplist_allmodels[[j]] <- toadd
  }
}

```

```

    print(models2[j])
  }

  saveRDS(emplist,
    paste0("__data/timeseries/sst_year_proj_ens_",
      ssp, ".RDS"))
  saveRDS(emplist_allmodels,
    paste0("__data/timeseries/sst_year_proj_ind_",
      ssp, ".RDS"))
}

ssps <- c("ssp245", "ssp585")
tic(); future_walk(ssps, timeseries_ssp); toc() #26 seconds for both

# For loop version
#tic(); timeseries_ssp("ssp245"); toc() #Jessie: 16.32 seconds for one
#timeseries_ssp("ssp585")

```

```

# Ensemble mean
ens <- readRDS("__data/timeseries/sst_year_proj_ens_ssp245.RDS")
ens <- do.call (rbind, ens)
head(ens)

```

```

# For each ESM...
ind <- readRDS("__data/timeseries/sst_year_proj_ind_ssp245.RDS")
ind <- do.call (rbind, ind)
head(ind)

```

Caution

Today, we're using the ensemble mean. But in your own work, you may want to consider using the ensemble median instead. The median may be less impacted by outliers/extremes in the data, especially if you're using ESMs that are known to be 'too-hot' in your region of interest.

7.1.2 Bind everything and apply 5-yr smooth

```
smooth_esms <- function(ssp, window_size) {  
  
  # Ensemble mean  
  timeseries_proj_ens <- readRDS(paste0("__data/timeseries/sst_year_proj_ens_", ssp, ".RDS"))  
  
  ens <- do.call(rbind, timeseries_proj_ens)  
  zoo_data <- zoo(ens$value, order.by = ens$date)  
  smoothed_esm <- rollapply(zoo_data, width = window_size,  
                           FUN = mean,  
                           align = "center",  
                           fill = "extend")  
  smooth_esm <- data.frame(date = time(smoothed_esm),  
                           values = coredata(smoothed_esm))  
  assign(paste0("smooth_esm_", ssp),  
        smooth_esm,  
        envir = globalenv())  
  
  # Individual models  
  timeseries_proj <- readRDS(paste0("__data/timeseries/sst_year_proj_ind_", ssp, ".RDS"))  
  timeseries_allmodels <- do.call(rbind, timeseries_proj)
```

```

allmodels <- timeseries_allmodels
emplist <- list()
ens2 <- allmodels

for (i in 1:length(unique(allmodels$model))) {

  ens <- subset(ens2, model == unique(ens2$model)[i])
  zoo_data <- zoo(ens$value, order.by = ens$date)
  smoothed_esm <- rollapply(zoo_data, width = window_size,
                           FUN = mean, align = "center", fill = "extend")
  smooth_11_esm <- data.frame(date = time(smoothed_esm),
                              values = coredata(smoothed_esm))
  smooth_11_esm$model <- unique(ens$model)
  emplist[[i]] <- smooth_11_esm
}

ssp_smoothed_ind <- do.call(rbind, emplist)
assign(paste0("ssp_smoothed_ind_", ssp),
      ssp_smoothed_ind,
      envir = globalenv())
}

tic(); smooth_esms("ssp245", window_size = 5); toc()
smooth_esms("ssp585", window_size = 5)

```

7.1.3 Plot!

```

plot_ts <- function(ssp, sspletter) {

  smooth_esm <- get(paste0("smooth_esm_", ssp)) #relies on this being in global env
  ssp_smoothed_ind <- get(paste0("ssp_smoothed_ind_", ssp))

  # Make a kick-ass plot

```

```

p1 <- ggplot() +
  geom_line(smooth_esm,
    mapping = aes(x = date, y = values),
    lwd = 1.5) +
  geom_rect(data = data.frame(),
    mapping = aes(xmin = 2020, xmax = 2040, ymin = -Inf, ymax = Inf),
    fill = "grey",
    alpha = 0.4) +
  geom_rect(data = data.frame(),
    mapping = aes(xmin = 2080, xmax = 2100, ymin = -Inf, ymax = Inf),
    fill = "grey",
    alpha = 0.4) +
  geom_line(smooth_esm,
    mapping = aes(x = date, y = values),
    lwd = 1.5) +
  geom_line(subset(ssp_smoothed_ind, model == "ACCESS-CM2"),
    mapping = aes(x = date, y = values),
    col = "black",
    alpha = 0.3) +
  geom_line(subset(ssp_smoothed_ind, model == "IPSL-CM6A-LR"),
    mapping = aes(x = date, y = values),
    col = "black",
    alpha = 0.3) +
  theme_bw() +
  scale_x_continuous(name = "Year",
    n.breaks = 6) +
  scale_y_continuous(name = "SST (°C)",
    limits = c(13.5, 20.5)) +
  theme(panel.grid.minor = element_blank(),
    plot.margin=unit(c(1,0.1,.1,0.1),"cm"),
    axis.title = element_text(size = 20,
      family = "Arial Narrow",
      face = "bold"),
    axis.text = element_text(size = 20,
      family = "Arial Narrow"),
    axis.title.x = element_text(margin = margin(t = 10, r = -20))) +
  annotate("text", x = 2019, y = 20.1,
    label = sspletter,
    size = 9,
    fontface = "bold",
    family = "Arial Narrow",
    hjust = 0,

```

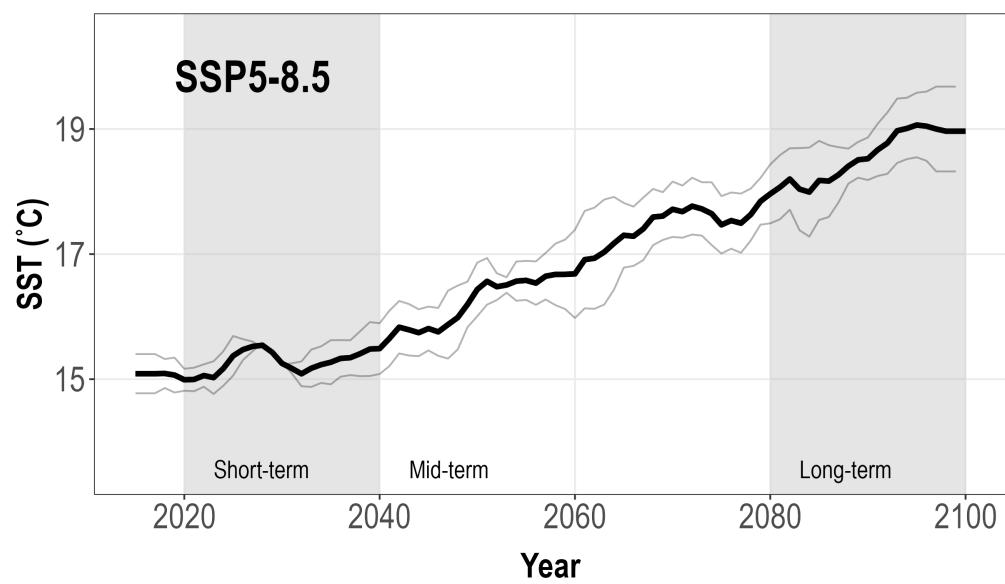
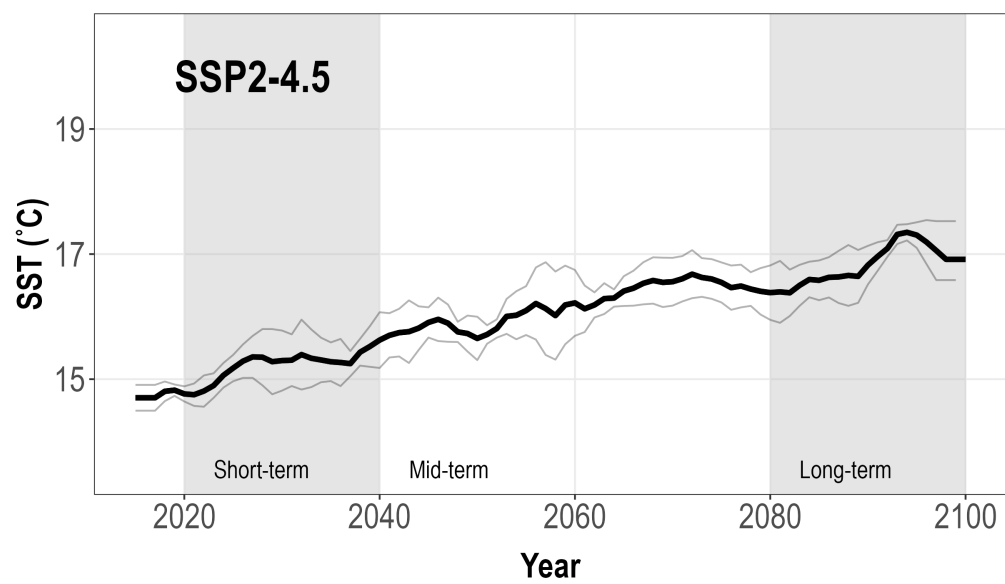
```

        vjust = 1) +
  annotate("text", x = 2083, y = 13.7,
    label = "Long-term", size = 5,
    family = "Arial Narrow",
    hjust = 0,
    vjust = 1) +
  annotate("text", x = 2023, y = 13.7,
    label = "Short-term", size = 5,
    family = "Arial Narrow",
    hjust = 0,
    vjust = 1)

  ggsave(p1,
    filename = paste0(outdir, "/",
                      ssp, "_SST_timeseries_1995-2100_11yrsmooth.png"),
    width = 8, height = 5)
}

# Run function
tic(); plot_ts("ssp245", sspletter = "SSP2-4.5"); toc() #Jessie: 0.197 seconds
plot_ts("ssp585", sspletter = "SSP5-8.5")

```

7.2 Projections

7.2.1 Individual ESM projections

```
models = c("ACCESS-CM2", "IPSL-CM6A-LR")
ssps = c("ssp245", "ssp585")
term = c("near", "mid", "long")
outdir = paste0(pth, "/_data/projections")
indir_proj <- paste0(pth, bc_pth, bc_pth_bc)

termdf <- data.frame(timeperiod = c("near", "long"),
                     st = c("2020-01-01", "2080-01-01"),
                     fin = c("2040-01-01", "2100-01-01"))

tic(); for (k in term) {
  for (i in ssps) {
    for (j in models) {

      allfiles_proj <- list.files(indir_proj, pattern = j, full.names = T)
      allfiles_proj <- allfiles_proj[grepl(i, allfiles_proj)]
      allfiles_proj <- allfiles_proj[grepl("2100", allfiles_proj)]
      rr <- rast(allfiles_proj)

      # subset to time period
      tp <- subset(termdf, timeperiod == k)
      rr <- rr[[time(rr) > tp[, "st"] & time(rr) < tp[, "fin"] ]]

      # Mean and SD
      proj_u <- mean(rr)
      proj_sd <- stdev(rr)
      # write to outdir
      filename_u <- paste0(outdir, "/ind/mean_", j, "_", i, "_", k, "_", "proj.nc" )
      filename_sd <- paste0(outdir, "/ind/sd_", j, "_", i, "_", k, "_", "proj.nc" )
      terra::writeCDF(proj_u, filename_u, overwrite = T)
      terra::writeCDF(proj_sd, filename_sd, overwrite = T)
    }
  }
}; toc() #Jessie: 1.8 seconds
```

```
list.files(paste0(pth, "/__data/projections/ind"))
```

7.2.2 Ensembled projections

```
tic(); for (k in term) {  
  for (i in ssps) {  
  
    allfiles_proj <- list.files(paste0(pth, "/__data/projections/ind"),  
                                pattern = i, full.names = T)  
    allfiles_proj <- allfiles_proj[grepl(k, allfiles_proj)]  
    allfiles_proj <- allfiles_proj[grepl("mean", allfiles_proj)]  
    rr <- rast(allfiles_proj)  
  
    # Mean and SD  
    proj_u <- mean(rr)  
    proj_sd <- stdev(rr)  
  
    # write to outdir  
    filename_u <- paste0(outdir, "/ens/mean_ens_", i, "_", k, "_", "proj.nc" )
```

```

    filename_sd <- paste0(outdir, "/ens/sd_ens_", i, "_", k, "_", "proj.nc" )
    terra::writeCDF(proj_u, filename_u, overwrite = T)
    terra::writeCDF(proj_sd, filename_sd, overwrite = T)
  }
}; toc() #Jessie: 0.3 seconds

```

```
list.files(paste0(pth, "/__data/projections/ens"))
```

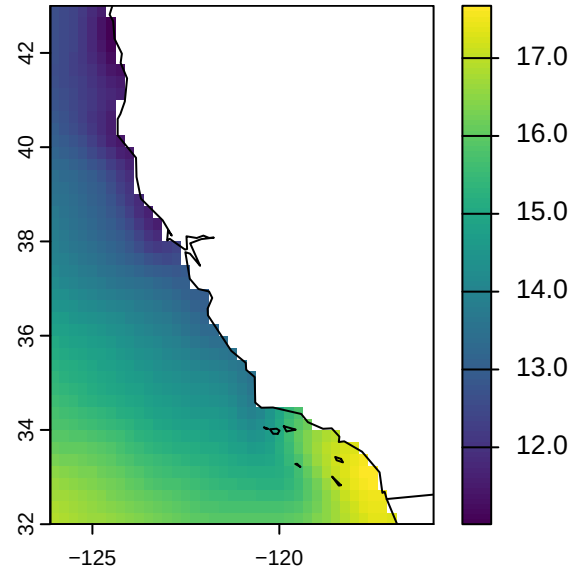
7.2.3 Delta difference

```

#Ensemble average of 1995-2014 for both models
bc_pth <- paste0(pth, "/__data/bias_correct/esm/_4_bias_corrected/")
bc_hist_remap_access <- rast(paste0(bc_pth, "tos_mo_ACCESS-CM2_1995-2014_bc_historical_remap")
bc_hist_remap_ipsl <- rast(paste0(bc_pth, "tos_mo_IPSL-CM6A-LR_1995-2014_bc_historical_remap")
bc_hist_avg <- c(bc_hist_remap_access, bc_hist_remap_ipsl)
mean_hist <- mean(bc_hist_avg)
plot(mean_hist, main = "Ensembled SST 1995-2014"); maps::map("world", add = T)

```

Ensembled SST 1995–2014



```
ssps = c("ssp245", "ssp585")

tic(); for (i in ssps) {

  near_mean <- rast(paste0(pth, "__data/projections/ens/mean_ens_", i, "_near_proj.nc"))
  delta_near <- near_mean - mean_hist
  writeCDF(delta_near, paste0(outdir, "/delta_mean_ens_near_", i, ".nc"),
            overwrite = T)

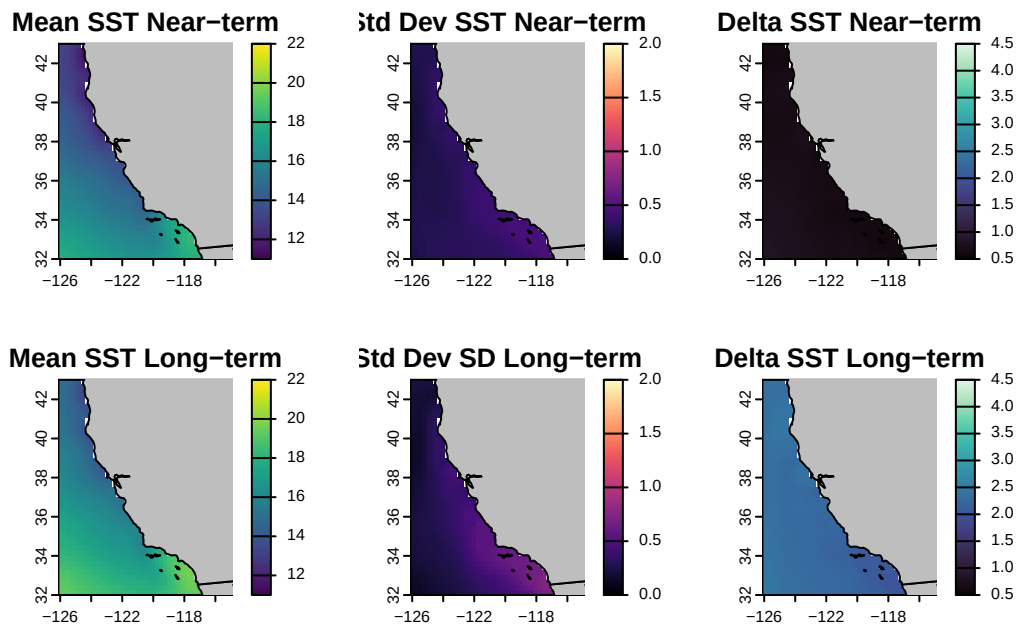
  mid_mean <- rast(paste0(pth, "__data/projections/ens/mean_ens_", i, "_mid_proj.nc"))
  delta_mid <- mid_mean - mean_hist
  writeCDF(delta_mid, paste0(outdir, "/delta_mean_ens_mid_", i, ".nc"),
            overwrite = T)

  long_mean <- rast(paste0(pth, "__data/projections/ens/mean_ens_", i, "_long_proj.nc"))
  delta_long <- long_mean - mean_hist
  writeCDF(delta_long, paste0(outdir, "/delta_mean_ens_long_", i, ".nc"),
            overwrite = T)

}; toc()
```

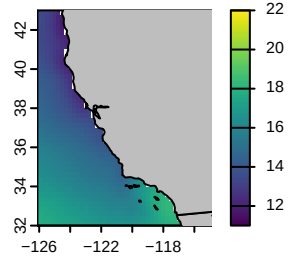
```
list.files(paste0(pth, "/__data/projections/delta"))
```

7.2.4 Plot: SSP2-4.5

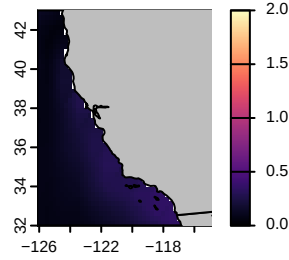


7.2.5 Plot: SSP5-8.5

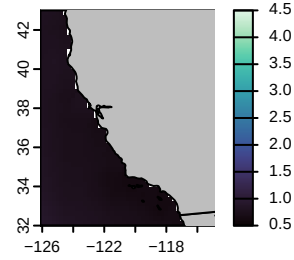
Mean SST Near-term



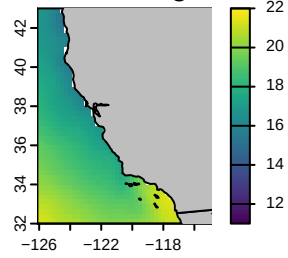
Std Dev SST Near-term



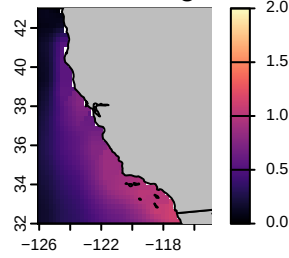
Delta SST Near-term



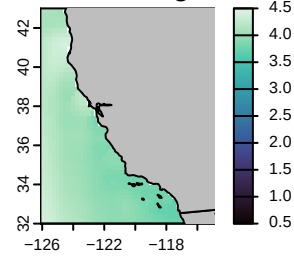
Mean SST Long-term



Std Dev SD Long-term



Delta SST Long-term



7.3 Uncertainty

8 Handy resources

- [CDO reference card](#)
- [hotrstuff](#)
- [climate4R](#)
- [CDO explainer](#) [CDO explainer](#)
- [CDO news article](#)

References

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Feedback and contributing

[Github repository](#)

